1. Probability Distributions

1.1. Overview

Definition 1.1. A random variable is a function that is defined over a sample space.

Definition 1.2. The probability distribution for a discrete variable Y is a formula or a table, showing values of $p(y)$ for all y.

Theorem 1.3. For any discrete probability distribution, the following must be true:
1. $0 \leq p(y) \leq 1$ for all y.
2. $\sum_y p(y) = 1$ where the sum is over all values of y with nonzero probability.

These come from the Kolmogorov axioms.

1.2. Expected value and variance

Definition 1.4. For some discrete random variable Y, with probability function $p(y)$,

$$E(Y) = \sum_y yp(y) = \mu$$

The expected value is a probability-weighted average of all possible values of Y.

Theorem 1.5.

For a discrete random variable Y with probability function $p(y)$, and some function $g(Y)$ which is a real-valued function of Y, the expected value of $g(Y)$ is

$$E[g(Y)] = \sum_y g(y)p(y)$$

This means when we have something like $E(Y^2)$ we can calculate this as $\sum_y y^2 p(y)$.

Theorem 1.6. If Y is a random variable with mean $E(Y) = \mu$, the variance of Y is

$$\begin{aligned} V(Y) &= E[(Y-\mu)^2] \\ &= E(Y^2) - \mu^2 \end{aligned}$$

As usual, the standard deviation of Y is the positive square root of the variance.

2. The Binomial Distribution

2.1. Overview

A binomial experiment has the following properties:
1. A fixed number of identical trials
2. Two possible outcomes for each trial (success or failure)
3. The probability of success for each trial is some constant value, p, and the probability of failure is $q = (1-p)$

4. The trials are independent
5. The random variable of interest is the number of successes observed out of n total trials

The sample points generated by a binomial experiment are collections of successes and failures, for a total of n trials: $\{SFSSFF\}, \{FFFFSS\}$, etc.

The total number of permutations of the above experiment ($n = 6$ trials) is $\binom{6}{y}$ for y successes. So if we want to find the number of permutations where there are 2 successes, we calculate the following:

$$\binom{6}{2} = \frac{6!}{2!4!} = \frac{6 * 5}{2} = 15$$

There are fifteen ways to have 4 failures and 2 successes. To get the probability of this happening, we need to multiply all of the independent probabilities of each of the trials. So:

$$P(Y = 2) = p(y) = 15p^2q^4$$

because there are 15 ways to have 4 failures and 2 successes, and for each of these, the probability of its occurrence is equal to the product of the probabilities of each independent event. There are 4 failures, each with probability q, and 2 successes, each with probability p.

Definition 2.1. In general, for a binomial experiment, the probability of y successes in n trials is:

$$p(y) = \binom{n}{y} p^y q^{n-y}$$

We can use R to compute binomial probabilities.

To find $P(Y = y)$ in n trials with success probability p:

```
dbinom(n, y, p)
```

To find $P(Y \leq y)$:

```
pbinom(n, y, p)
```

2.2. Expected value and variance

Theorem 2.2. For a binomial experiment with random variable of interest Y, success probability p, and n total trials:

$$E(Y) = np$$

$$V(Y) = npq$$

3. The Geometric Distribution

The geometric distribution is like the binomial distribution except the random variable of interest is the number of the trial on which the first success occurs. This means the sample points are all strings of failures that end with a success, like $\{FFFFFS\}$.

The probability of such an occurrence is q raised to the power of the number of failures, times p. This means that if we want to find the probability of success happening on the third trial, we calculate the following: q^2p.

Theorem 3.1. For a geometric distribution with a random variable Y, probability of success p, and probability of failure q,

$$p(y) = q^{y-1}p$$

$$E(Y) = \frac{1}{p}$$

$$V(Y) = \frac{1-p}{p^2}$$

4. The Hypergeometric Distribution

The hypergeometric distribution is used for experiments where the we are selecting a sample of a size that is large relative to the population size, and sampling occurs without replacement. The total number of sample points if we draw a random sample of size n from a population of size N is $\binom{N}{n}$. The number of successes in the population is r.

Definition 4.1. For a random variable Y with a hypergeometric probability distribution,

$$p(y) = \frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}}$$

This is because there are r successes, from which we choose y, and $N-r$ failures, from which we choose $n-y$. This gives us the number of ways to choose a sample of size n from a population of size N with y successes. We put this over $\binom{N}{n}$, the total number of ways to choose a sample of size n.

4.1. Expected value and variance

Theorem 4.2.

$$E(Y) = \frac{nr}{N} = np$$

because p is equal to $\frac{r}{N}$. And

$$V(Y) = n\left(\frac{r}{N}\right)\left(\frac{N-r}{N}\right)\left(\frac{N-n}{N-1}\right) = npq\left(\frac{N-n}{N-1}\right)$$

where $\frac{N-n}{N-1}$ is a finite population correction.

Note that as $N \to \infty$, the formula for the variance of a hypergeometric random variable approaches that of a geometric variable.